

Exercises1

In the following find the indicated integral. Check your answer by differentiation.

1. $\int x^5 dx$
2. $\int x^{\frac{3}{4}} dx$
3. $\int \sqrt{x} dx$
4. $\int 5 dx$
5. $\int 3e^x dx$
6. $\int \frac{1}{x^2} dx$
7. $\int (3x^2 - \sqrt{5x} + 2) dx$
8. $\int (\frac{e^x}{2} + x\sqrt{x}) dx$
9. $\int (2e^x + \frac{6}{x} + \ln 2) dx$
10. $\int \frac{x^2 + 3x - 2}{\sqrt{x}} dx$
11. $\int x^3 (2x + \frac{1}{x}) dx$
12. $\int x(2x + 1)^2 dx$
13. $\int \sqrt{x}(x^2 - 1) dx$.

Exercises2

In Exercises 1 – 13, find the indefinite integrals.

1. $\int x \sin x dx$
2. $\int x^2 \sin x dx$
3. $\int t^3 \sin t dt$
4. $\int x \ln x^2 dx$
5. $\int \sqrt{t} \ln t dt$
6. $\int (\ln x)^2 dx$
7. $\int x^2 \ln x dx$
8. $\int x^3 \ln x dx$
9. $\int x^n \ln x dx$, where n is a positive integer
10. $\int x^n \ln x^m dx$, where m and n are positive integers
11. $\int \sin(\ln x) dx$

(Hint: Let $u = \sin(\ln x)$, and do integration by parts twice)

12. $\int \cos(\ln x) dx$

13. $\int x \ln(x+1) dx$ (Hint: Take $u = \ln(x+1)$,

$dv = x dx$, and $v = (x^2 - 1)/2$.)

Exercises 3

In Exercises 1 – 12 evaluate the given integral by making the indicated substitution.

1. $\int \cos \pi x \, dx$; $u = \pi x$

2. $\int_0^{\pi/6} 2 \sin(-3x) \, dx$; $u = -3x$

3. $\int_1^8 \sqrt{3t+1} \, dt$; $u = 3t+1$

4. $\int (1-4x^2)^{5/2} (8x) \, dx$; $u = 1-4x^2$

5. $\int 2x \sin x^2 dx$; $u = x^2$

6. $\int \frac{6x+5}{3x^2+5x} dx$; $u = 3x^2+5x$

7. $\int \frac{6x+5}{(3x^2+5x)^3} dx$; $u = 3x^2+5x$

8. $\int \frac{1}{y \log y} dy$; $u = \log y$

9. $\int x \sqrt{x-1} \, dx$; $u = x-1$

10. $\int x^2 \sqrt{x+2} \, dx$; $u = x+2$

11. $\int \sec x \tan x \sqrt{3+\sec x} \, dx$; $u = 3+\sec x$

$$12. \int \frac{\sqrt[3]{x}}{\sqrt[3]{x+1}} dx ; u = \sqrt[3]{x+1}$$

In Exercises 13 – 28 evaluate the given integral.

$$13. \int \sqrt{4x+5} \, dx$$

$$14. \int \sqrt{6-2x} \, dx$$

$$15. \int_0^{\frac{1}{4}} \sec^2 \pi x \, dx$$

$$16. \int t^6 (1-4t^7)^{\frac{8}{3}} dt$$

$$17. \int t^6 (1-4t^7)^8 dt$$

$$18. \int (s^2 - 3)^{19} s \, ds$$

$$19. \int (4x+3)(2x^2+3x+4)^4 dx$$

$$20. \int_0^3 (z-3)(z^2-6z+9)^5 dz$$

$$21. \int x\sqrt{3x^2+5} \, dx$$

$$22. \int x^3 (x^4+5)^{\frac{1}{3}} dx$$

$$23. \int t\sqrt{2t-1} \, dt$$

$$24. \int \frac{x^2}{\sqrt{x+1}} dx$$

$$25. \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos^2 x} dx$$

$$26. \int x \left(\sqrt{x^2+1} + \frac{1}{\sqrt{x^2+1}} \right) dx$$

$$27. \int \frac{\sec^2(\ln x)}{x} dx$$

$$28. \int (\tan x) \sec^2[\ln(\cos x)] \, dx$$

$$29. \text{Find } \int_0^1 \ln(x+1) \, dx$$

$$30. \text{Find } \int \cos \sqrt{t} \, dt$$

$$31. \text{Find } \int x \sin ax \, dx, \text{ where } a \text{ is a nonzero constant.}$$

Exercises .4

Evaluate the given integral in each of the following.

$$1. \int \sin^3 x \cos^2 x \, dx$$

$$2. \int \sin^3 x \cos^3 x \, dx$$

$$3. \int \sin^3 3x \cos 3x \, dx$$

$$4. \int_0^{\pi/2} \sin^2 t \cos^5 t \, dt$$

$$5. \int \left(\frac{1}{x^2} \right) \sin^2 \frac{1}{x} \cos^5 \frac{1}{x} \, dx$$

$$6. \int \sin^2 x \cos^2 x \, dx$$

$$7. \int_0^{\pi} \sin^2 y \cos^4 y \, dy$$

$$8. \int_0^{\pi/2} \sin^2 2y \cos^2 2y \, dy$$

$$9. \int \sin^{10} x \cos^3 x \, dx$$

$$10. \int \sin^{-10} x \cos^3 x \, dx$$

$$11. \int \sin^6 w \cos^4 w \, dw$$

$$12. \int \sin^{-17} x \cos^3 x \, dx$$

$$13. \int (1 + \sin^2 x)(1 + \cos^2 x) \, dx$$

$$14. \int \tan^5 x \sec^2 x \, dx$$

$$15. \int \tan^3 5x \sec^2 5x \, dx$$

$$16. \int_0^{\pi/4} \tan^5 t \sec^4 t \, dt$$

$$17. \int x \tan^3 x^2 \sec^4 x^2 \, dx$$

$$18. \int_{5\pi/4}^{4\pi/3} \tan^3 x \sec x \, dx$$

$$19. \int \tan x \sec^3 x \, dx$$

$$20. \int \frac{1}{\sqrt{x}} \tan \sqrt{x} \sec^3 \sqrt{x} \, dx$$

$$21. \int \tan^2(1-y) \sec(1-y) \, dy$$

$$22. \int \tan x \sec^2 x \, dx$$

$$23. \int_0^{\pi/3} \tan x \sec^2 x \, dx$$

$$24. \int \tan x \sec^5 x \, dx$$

$$25. \int \csc x \, dx$$

$$26. \int \csc^3 x \, dx$$

$$27. \int \cot^3 x \csc^2 x \, dx$$

$$28. \int \cot^3 t \csc^4 t \, dt$$

$$29. \int_{\pi/4}^{\pi/2} \cot^3 x \csc^3 x \, dx$$

$$30. \int_{\pi/3}^{\pi/4} \cot x \csc^3 x \, dx$$

$$31. \int \cot t \csc^{-2} t \, dt$$

$$32. \int \frac{\cot t}{\csc^3 t} dt$$

$$33. \int \frac{\tan^3 t}{\sec^4 t} dt$$

$$34. \int \tan^3 x \csc^2 x \, dx$$

$$35. \int_{\pi/4}^{\pi/2} \frac{1}{1 + \cos x} dx$$

$$36. \int \frac{1 + \cos x}{\sin x} dx$$

$$37. \int \sin ax \sin bx \, dx$$

(Hint :Use the identity $\sin x \sin y = \frac{1}{2} \cos(x - y) - \frac{1}{2} \cos(x + y)$.)

$$38. \int \cos ax \cos bx \, dx$$

(Hint :Use the identity $\cos x \cos y = \frac{1}{2} \cos(x + y) + \frac{1}{2} \cos(x - y)$.)

integrals that commonly occur in applications.

Exercises.5

Evaluate the integral in each of the following:

$$1. \int_0^{1/2} \sqrt{1 - 4x^2} \, dx$$

$$2. \int_0^4 \sqrt{16 - x^2} \, dx$$

$$3. \int_{-2}^2 \sqrt{1 - \frac{x^2}{4}} \, dx$$

$$4. \int \sqrt{1 + (2x - 1)^2} \, dx$$

$$5. \int_{-\sqrt{2}}^{\sqrt{2}} \frac{1}{\sqrt{4 - x^2}} dx$$

$$6. \int_0^{\sqrt{3}} \frac{t^5}{\sqrt{t^2 + 1}} dt$$

$$7. \int \frac{1}{(x^2 + 1)^{3/2}} dx$$

$$8. \int \frac{1}{(2x^2 + 1)^{3/2}} dx$$

$$9. \int_0^1 \frac{1}{(3x^2 + 2)^{5/2}} dx$$

$$10. \int \frac{1}{(x^2 - 4)^{3/2}} dx$$

$$11. \int \frac{1}{(9x^2 - 4)^{5/2}} dx$$

$$12. \int \frac{x^2}{(x^2 - 2)^{5/2}} dx$$

$$13. \int \frac{1}{(3 - x^2)^{3/2}} dx$$

$$14. \int \frac{1}{(1 - 2w^2)^{5/2}} dw$$

$$15. \int \frac{1}{\sqrt{25 - 4x^2}} dx$$

$$16. \int_0^{\sqrt{3}/2} \frac{x^2}{\sqrt{1 - x^2}} dx$$

$$17. \int_0^1 \frac{4}{x^2 + 1} dx$$

$$18. \int_1^{\sqrt{3}} \frac{1}{\sqrt{1 + x^2}} dx$$

$$19. \int_1^{\sqrt{2}} \frac{1}{\sqrt{2x^2 - 1}} dx$$

$$20. \int \frac{x^2}{\sqrt{9x^2 - 1}} dx$$

$$21. \int_{-1}^1 \frac{1}{t^2 + 2t + 5} dt \quad (\text{Hint : Complete the square.})$$

Exercises .6

Integrate.

$$1. \int \frac{1}{x^2 + x - 2} dx$$

$$2. \int \frac{x + 3}{x^2 + x} dx$$

$$3. \int \frac{x^3 + 5}{x^2 - 25} dx$$

$$4. \int \frac{x}{x^2 + 2x - 8} dx$$

$$5. \int \frac{(x + 2)}{x^3 + x^2 - 2x} dx$$

$$6. \int \frac{x^2 + 9}{x(x - 3)^2} dx$$

$$7. \int \frac{5x + 3}{x^3 - 2x^2 - 3x} dx$$

$$8. \int_0^1 \frac{x}{x^2 - 3x - 4} dx$$

$$9. \int_2^3 \frac{x+1}{x^2(x-1)} dx$$

$$10. \int \frac{x^3 + x^2 + 3}{x^2 + 3} dx$$

$$11. \int \frac{x^3 dx}{x^2 - 1}$$

$$12. \int \frac{2x-1}{x^2-1} dx$$

$$13. \int \frac{dx}{(x+1)(2x^2+3x+1)}$$

$$14. \int \frac{2x}{x^3-8} dx$$

$$15. \int \frac{x dx}{(x-1)^3}$$

$$16. \int \frac{dx}{x^2 - a^2}$$

$$17. \int \frac{dx}{x(1 + \ln x)^2}$$

$$18. \int_0^1 \frac{dx}{e^x + 1}$$

$$9. \int_0^4 \frac{1}{\sqrt{6y+1}} dy$$

$$10. \int_1^2 \frac{x^2}{(x^3+1)^2} dx$$

11. An object is moving so that its speed after t minutes is $5 + 2t + 3t^2$ meters per minute. How far does the object travel during the second minute?
12. At a certain factory, the marginal cost is $6(q-5)^2$ E.P. per unit when the level of production is q units. By how much will the total manufacturing cost increase if the level of production is raised from 10 to 13 units?

Exercise 7

In Exercises (1) – (4) compute the "Riemann sum for the given function on the indicated interval. Specify $\|p\|$.

- (1) $f(x) = x$, $[0, 4]$, two subintervals;

$$x_0 = 0, x_1 = \frac{5}{2}, x_2 = 4; x_1^* = 2, x_2^* = 3.$$

- (2) $f(x) = -2x$, $[-1, 2]$, three subintervals;

$$x_0 = -1, x_1 = -\frac{1}{2}, x_2 = 1, x_3 = 2; x_1^* = -\frac{1}{2}, x_2^* = 0, x_3^* = \frac{3}{2}.$$

- (3) $f(x) = \sin x$, $[0, 2\pi]$, three subintervals;

$$x_0 = 0, x_1 = \pi, x_2 = \frac{3\pi}{2}, x_3 = 2\pi, x_1^* = \frac{\pi}{2}, x_2^* = \frac{7\pi}{6}, x_3^* = \frac{7\pi}{4}$$

- (4) $f(x) = \cos x$, $[-\frac{\pi}{2}, \frac{\pi}{2}]$, for subintervals;

$$x_0 = -\frac{\pi}{2}, x_1 = -\frac{\pi}{4}, x_2 = 0, x_3 = \frac{\pi}{3}, x_4 = \frac{\pi}{2};$$

$$x_1^* = -\frac{\pi}{3}, x_2^* = -\frac{\pi}{6}, x_3^* = \frac{\pi}{4}, x_4^* = \frac{\pi}{3}$$

In exercises (5) – (10) evaluate the given definite integral

$$(5) \int_1^0 3dx$$

$$(6) \int_{-1}^4 (-2)dx$$

$$(7) \int_{-3}^1 x dx$$

$$(8) \int_0^3 x dx$$

$$(9) \int_1^2 (x^2 - x) dx$$

$$(10) \int_0^1 (x^3 - 1) dx$$

- (11) Use the BASIC program of this section to obtain approximate values

for $\int_0^1 \sqrt{x^3 + 1} dx$ by choosing $n = 500$ a

Exercises8

Use calculus to find the area of the region R in the following.

1. R is the triangle bounded by the line $y = 4 - 3x$ and the coordinate axes.
2. R is the triangle with vertices $(-4,0)$, $(2,0)$ and $(2,6)$.
3. R is the rectangle with vertices $(1,0)$, $(-2,0)$, $(-2,5)$ and $(1,5)$.
4. R is the region bounded by the curve $y = 4x^3$, the line $x = 2$ and the x axis.
5. R is the region bounded by the curve $y = 1 - x^2$ and the x axis.
6. R is the region in the first quadrant bounded by the curve $y = 4 - x^2$ and the lines $y = 3x$ and $y = 0$.